

Name: _____ Class: _____

WHITEBRIDGE HIGH SCHOOL



2008

Trial HSC Examination (Assessment 4)

MATHEMATICS EXTENSION 2

Time Allowed: 3 hours
(plus 5 minutes reading time)

Directions to Candidates

- All questions of equal value.
- Commence each question on a new page.
- Marks may be deducted for careless or badly arranged work.

Question 1 (15 marks) Commence each question on a SEPARATE page

a. Evaluate $\int_0^3 \frac{x}{\sqrt{16+x^2}} dx$. **3**

b. Find $\int \frac{dx}{x^2 + 6x + 13}$ **2**

c. Find $\int xe^{-x} dx$. **2**

d. Find $\int \cos^3 \theta d\theta$. **3**

e. (i) Find constants A , B and C such that **3**

$$\frac{x^2 - 4x - 1}{(1 + 2x)(1 + x^2)} \equiv \frac{A}{1 + 2x} + \frac{Bx + C}{1 + x^2}$$

(ii) Hence find $\int \frac{x^2 - 4x - 1}{(1 + 2x)(1 + x^2)} dx$. **2**

Question 2 (15 marks) Commence each question on a SEPARATE page

a. Given that $z = 1 + i$ and $w = -3$, find, in the form $x + iy$:

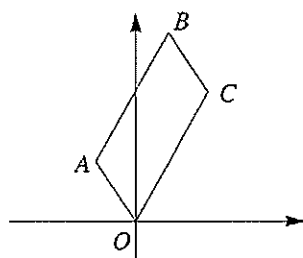
(i) wz^2 **1**

(ii) $\frac{z}{z + w}$ **2**

b. Using de Moivre's theorem, simplify $(-1 - i\sqrt{3})^{-10}$, expressing the answer in the form $x + iy$. **3**

c. Sketch the region described by the following $|z| < 2$ and $\frac{2\pi}{3} \leq \arg z \leq \frac{5\pi}{6}$. **2**

d.

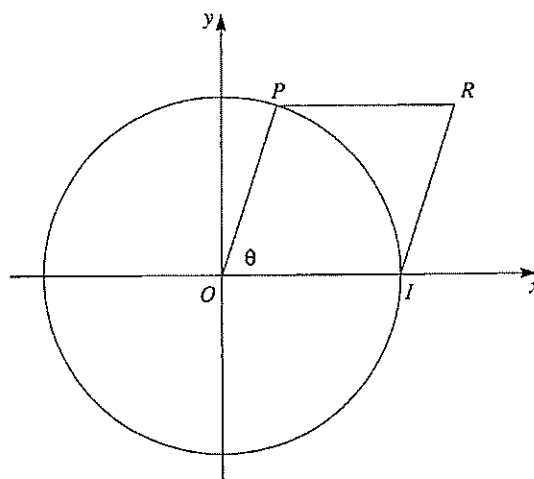


In the diagram above, $OABC$ is a parallelogram with $OA = \frac{1}{2} OC$. **3**

The point A represents the complex number $-\frac{1}{2} + i\frac{\sqrt{3}}{2}$.

If $\angle AOC = 60^\circ$, what complex number does C represent?

- e. In the Argand diagram below, P represents $\cos \theta + i \sin \theta$, I represents the number $1 + 0i$, and R represents the number $z = 1 + \cos \theta + i \sin \theta$.



- (i) Using the properties of the rhombus, or otherwise, show that z can be expressed as $z = 2\cos \frac{\theta}{2}(\cos \frac{\theta}{2} + i \sin \frac{\theta}{2})$. **2**
- (ii) Hence show that $\frac{1}{z} = \frac{1}{2} - \frac{i}{2} \tan \frac{\theta}{2}$. **2**

Question 3 (15 marks) Commence each question on a SEPARATE page

a. Let $1, \omega, \omega^2$ be the three cube roots of unity.

(i) Show that:

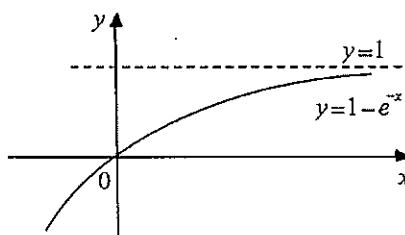
(α) $\omega^3 = 1$ **1**

(β) $1 + \omega + \omega^2 = 0$ **1**

(ii) If $1, \omega, \omega^2$ are the roots of $x^3 + ax^2 + bx + c = 0$, find a, b and c . **3**

b. Find the acute angle between the tangent $x^3 + y^3 = 1$ at $x = 1$ and the line $y = x$. **4**

c. The graph shows the graph of $f(x) = 1 - e^{-x}$. On separate diagrams, sketch the graphs of the following functions, showing clearly the equations of any asymptotes:



(i) $y = [f(x)]^2$ **2**

(ii) $y = \frac{1}{f(x)}$ **2**

(iii) $y = \sqrt{f(x)}$ **2**

Question 4 (15 marks) Commence each question on a SEPARATE page

- a. Show that the roots of the equation $z^{10} = 1$ are given by $z = \cos \frac{r\pi}{5} + i \sin \frac{r\pi}{5}$ **2**

where $r = 0, 1, 2, 3, \dots, 9$

- b. The equation $x^3 + 2x + 1 = 0$ has roots α, β and γ .

- (i) Find the monic cubic equation with roots $\frac{1}{\alpha}, \frac{1}{\beta}$ and $\frac{1}{\gamma}$. **2**

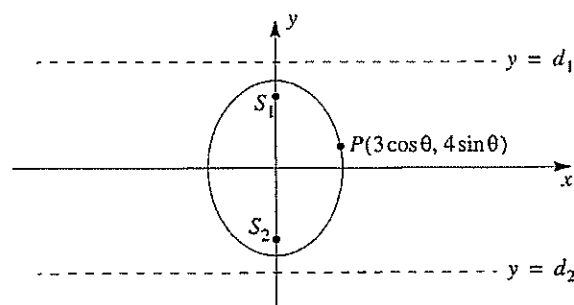
- (ii) Find the monic cubic equation with roots $\frac{\beta + \gamma}{\alpha^2}, \frac{\gamma + \alpha}{\beta^2}$ and $\frac{\alpha + \beta}{\gamma^2}$. **3**

- c. Given that $x = \theta + \frac{1}{2}\sin 2\theta$ and $y = \theta - \frac{1}{2}\sin 2\theta$:

- (i) Show that $\frac{dy}{dx} = \tan^2 \theta$ **2**

- (ii) Show that $\frac{d^2y}{dx^2} = \tan \theta \sec^4 \theta$ **2**

- d.



The diagram above shows an ellipse with parametric equation

$$x = 3 \cos \theta$$

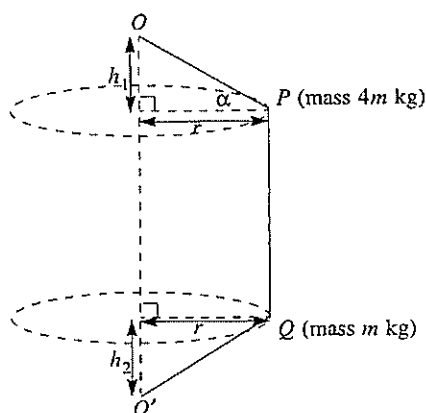
$$y = 4 \sin \theta$$

- (i) Write down the cartesian equations of the ellipse. **1**
- (ii) Find the coordinates of the foci S_1 and S_2 . **2**
- (iii) Find the equation of the directrices $y = d_1$ and $y = d_2$. **2**

Question 5 (15 marks) Commence each question on a SEPARATE page

- a. (i) Let $P(x)$ be a polynomial of degree 4 with a zero of multiplicity 3. **2**
Show that $P'(x)$ has a zero of multiplicity 2.
- (ii) Hence, or otherwise, find all zeros of $P(x) = 8x^4 - 25x^3 + 27x^2 - 11x + 1$, **2**
given that it has a zero of multiplicity 3.
- (iii) Sketch $y = 8x^4 - 25x^3 + 27x^2 - 11x + 1$, clearly showing the intercepts **1**
on the coordinate axes.
Do NOT give the coordinates of turning points and inflections.
- b. The roots of $a \tan^2 \alpha + b \tan \alpha + c = 0$ are $\tan \alpha_1$ and $\tan \alpha_2$. **2**
Show that the value of $\tan(\alpha_1 + \alpha_2)$ is $\frac{b}{c - a}$.
- c. A particle moves in a circle of radius r , with a constant speed rw . **1**
Write down the magnitude and direction of its acceleration.

d.



The diagram above shows two particles, P and Q, of masses $4m$ kg and m kg respectively, which are attached by a light inextensible string. The ends of the strings are attached to fixed points O and O'. O is vertically above O'. The particles P and Q move in horizontal circles, of equal radius r metres, about OO', with the same constant angular velocity w , so that Q always remains vertically above P.

The depth of P below the level of O is h_1 and the height of Q above the level of O' is h_2 . The angle that OP makes with the horizontal is α .

- (i) Let the tension in the string PQ be T newtons and the tension in the string OP be T_1 newtons. **2**

By drawing a force diagram and resolving forces acting on P, show that

$$T_1 \sin \alpha = 4mg + T$$

$$T_1 \cos \alpha = 4mw^2r$$

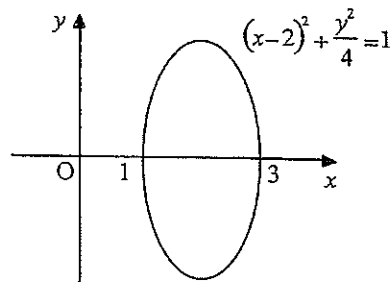
- (ii) Hence show that $h_1 = \frac{4mg + T}{4mw^2}$. **2**

- (iii) Hence show that $(4h_1 - h_2)w^2 = 5g$. **3**

Question 6 (15 marks) Commence each question on a SEPARATE page

- a. When the polynomial $P(x)$ is divided by $(x + 2)(x - 3)$ the remainder is $4x + 1$. **2**
What is the remainder when $P(x)$ is divided by $(x + 2)$?

b.



The region enclosed by the ellipse $(x - 2)^2 + \frac{y^2}{4} = 1$ is rotated through one complete revolution about the y -axis.

- (i) Use the method of cylindrical shells to show that the volume V of the solid of revolution is given by $V = 8\pi \int_1^3 x\sqrt{1 - (x - 2)^2} dx$ **2**
- (ii) Hence find the volume of the solid of revolution in simplest exact form. **3**
- c. From a point on the ground an object of mass m is projected vertically upwards with an initial speed of u . The object reaches a maximum height of H before falling back to the ground. The resistance due to air is equal to mkv^2 , and g is the acceleration due to gravity.
- (i) By using $\ddot{x} = v \frac{dv}{dx}$, show that $H = \frac{1}{2k} \ln \left(\frac{g + ku^2}{g} \right)$. **3**
- (ii) P is the point of height h above the point of projection. **2**
Let V be the speed of the object at P on its upward path.
Show that $h = \frac{1}{2k} \ln \left(\frac{g + ku^2}{g + kV^2} \right)$.
- (iii) During the downward path of the object it passes through P with half the speed of when it was first at P . **3**
Show that $V = \sqrt{\frac{3g}{k}}$.

Question 7 (15 marks) Commence each question on a SEPARATE page

- a. Find all the complex numbers $z = a + ib$, where a and b are real, such that $|z|^2 + 5\bar{z} + 10i = 0$. **2**

- b. Consider the rectangular hyperbola $xy = 4$.

(i) Show that the gradient of the tangent at the point $P(2p, \frac{2}{p})$ is $-\frac{1}{p^2}$. **2**

(ii) Show that the normal at P is given by $p^3x - py = 2(p^4 - 1)$. **2**

(iii) This normal meets the hyperbola again at $Q(2q, \frac{2}{q})$. **3**

By considering the product of the roots of the equation formed by the intersection of $xy = 4$ and $p^3x - py = 2(p^4 - 1)$, or otherwise, prove that $p^3q = -1$.

c. (i) Show that $\frac{t^n}{1+t^2} = t^{n-2} - \frac{t^{n-2}}{1+t^2}$. **2**

(ii) Let $I_n = \int \frac{t^n}{1+t^2} dt$. **1**

Show that $I_n = \frac{t^{n-1}}{n-1} - I_{n-2}$ for $n \geq 2$.

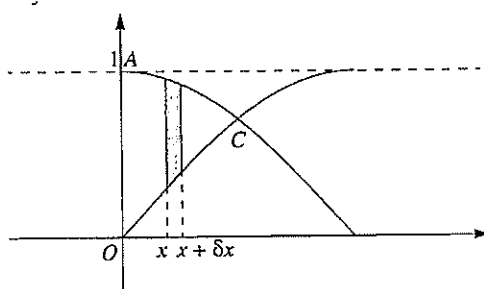
(iii) Show that $\int_0^1 \frac{t^6}{1+t^2} dt = \frac{13}{15} - \frac{\pi}{4}$. **3**

Question 8 (15 marks) Commence each question on a SEPARATE page

a. Find $\frac{dy}{dx}$ when $y = e^{xy}$.

2

- b. The diagram below shows part of the graphs of $y = \cos x$ and $y = \sin x$. The graph of $y = \cos x$ meets the y axis at A , and the C is the first point of intersection of the two graphs to the right of the y axis.



The region OAC is to be rotated about the line $y = 1$.

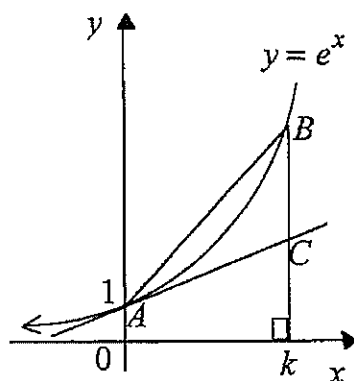
- (i) Write down the coordinates of the point C . **1**

- (ii) The shaded strip of width δx shown in the diagram is rotated about the line $y = 1$. Show that the volume δV of the resultant slice is given by **2**

$$\delta V = \pi(2\cos x - 2\sin x + \sin^2 x - \cos^2 x) \delta x.$$

- (iii) Hence evaluate the total volume when the region OAC is rotated about the line $y = 1$. **4**

c.



The curve $y = e^x$ cuts the y -axis at A .

B is a second point on the curve such that $x = k$ at B , where $k > 0$.

The tangent to the curve $y = e^x$ at A cuts the vertical line $x = k$ at the point C .

- (i) By considering areas, show that $\frac{1}{2}k(k+2) < e^k - 1 < \frac{1}{2}k(1+e^k)$. **3**

Hence deduce that $2.5 < e < 3$.

- (ii) Show that the curve $y = e^x$ bisects the area of $\triangle ABC$ for some value of k **3**

such that $2 < k < 3$. Taking $k = 2.7$ as a first approximation, apply Newton's method once to obtain a second approximation.

Give your answer to one decimal place.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1

$$a. \int_0^3 \frac{x}{\sqrt{16+x^2}} dx$$

$$\text{Let } u = 16+x^2$$

$$\frac{du}{dx} = 2x$$

$$dx = \frac{du}{2x}$$

$$= \int_{16}^{25} \frac{x}{\sqrt{u}} \cdot \frac{du}{2x}$$

$$= \frac{1}{2} \int_{16}^{25} u^{-\frac{1}{2}} du$$

$$= u^{\frac{1}{2}} \Big|_{16}^{25}$$

$$= 5 - 4$$

$$= 1$$

$$b. \int \frac{dx}{x^2+6x+13} = \int \frac{dx}{x^2+6x+9+4}$$

$$= \int \frac{dx}{4+(x+3)^2}$$

$$= \frac{1}{2} \tan^{-1} \frac{x+3}{2} + C$$

$$c. \int x e^{-x} dx = -x e^{-x} - \int 1 \cdot -e^{-x} dx$$

$$= -x e^{-x} + \int e^{-x} dx$$

$$= -x e^{-x} - e^{-x} + C$$

$$d. \int \cos^3 \theta d\theta = \int \cos^2 \theta \cos \theta d\theta$$

$$\text{Let } u = \sin \theta$$

$$\frac{du}{d\theta} = \cos \theta$$

$$d\theta = \frac{du}{\cos \theta}$$

$$= \int (1 - \sin^2 \theta) \cos \theta d\theta$$

$$= \int (1 - u^2) \cos \theta \cdot \frac{du}{\cos \theta}$$

$$= \int (1 - u^2) du$$

$$= u - \frac{u^3}{3} + C$$

$$= \sin \theta - \frac{\sin^3 \theta}{3} + C$$

$$e. i. \frac{x^2-4x-1}{(1+2x)(1+x^2)} = \frac{A(1+x^2)}{(1+2x)(1+x^2)} + \frac{(Bx+C)(1+2x)}{(1+2x)(1+x^2)}$$

$$\therefore x^2-4x-1 = A(1+x^2) + (Bx+C)(1+2x)$$

$$= A + Ax^2 + Bx + 2Bx^2 + C + 2Cx$$

$$= (A+2B)x^2 + (B+2C)x + A+C$$

$$\therefore A+2B=1 \quad \text{--- (1)}$$

$$B+2C=-4 \quad \text{--- (2)}$$

$$A+C=-1 \quad \text{--- (3)}$$

$$\text{From (1) } A=1-2B$$

$$\text{Subs in (3) } 1-2B+C=-1$$

$$\therefore 2B-C=2 \quad \text{--- (4)}$$

$$2 \times (4) \quad 4B-2C=4 \quad \text{--- (5)}$$

$$(2) + (5) \quad 5B=0$$

$$\therefore B=0$$

$$\text{Subs in (1) } A=1$$

$$\text{Subs in (3) } 1+C=-1$$

$$C=-2$$

$$\therefore A=1, B=0, C=-2$$

$$ii. \int \frac{1}{1+2x} + \frac{-2}{1+x^2} dx$$

$$= \frac{1}{2} \ln(1+2x) - 2 \tan^{-1} x + C$$

Question 2:

$$a. i. w z^2 = -3(1+i)^2$$

$$= -3(1+2i-1)$$

$$= -6i$$

$$ii. \frac{z}{z+w} = \frac{1+i}{-2+i} \times \frac{-2-i}{-2-i}$$

$$= \frac{-2-i-2i+1}{4+1}$$

$$= \frac{-1-3i}{5}$$

$$= -\frac{1}{5} - \frac{3i}{5}$$

$$b. (-1-i\sqrt{3})^{-10} \quad \text{Let } z = -1-i\sqrt{3}$$

$$|z|=2$$

$$\arg z = \tan^{-1} \frac{-\sqrt{3}}{-1}$$

$$= -\frac{2\pi}{3}$$

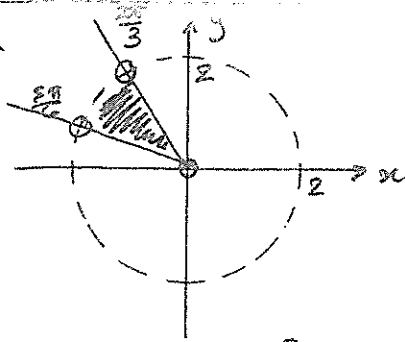
$$\therefore \left[2 \operatorname{cis} \left(-\frac{2\pi}{3} \right) \right]^{-10}$$

$$= \frac{1}{1024} \left[\cos \frac{20\pi}{3} + i \sin \frac{20\pi}{3} \right]$$

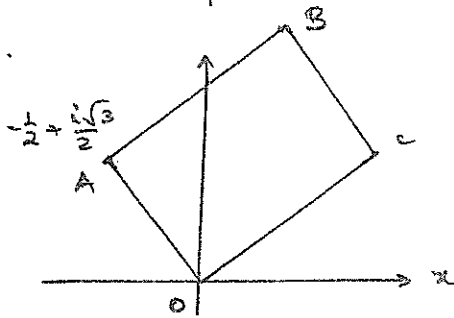
$$= \frac{1}{1024} \left[\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right]$$

$$= \frac{1}{1024} \left[-\frac{1}{2} + i \frac{\sqrt{3}}{2} \right]$$

$$= \frac{-1}{2048} + \frac{i\sqrt{3}}{2048}$$



d.



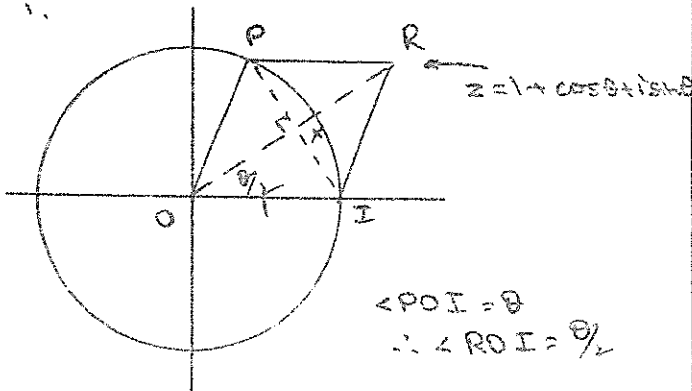
Now $AO = \frac{1}{2} OC$ and we rotate
 OA anticlockwise 60°
 i.e. multiply OA by $2 \times \text{cis } -60^\circ$

$$\therefore OC = 2 \times OA \times \text{cis } (-\pi/3)$$

Now, for OA mod: $\sqrt{\left(-\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$
 $= 1$
 $\arg = \tan^{-1} \frac{\frac{\sqrt{3}}{2}}{-\frac{1}{2}}$
 $= \frac{2\pi}{3}$

$$\begin{aligned}\therefore OC &= 2 \times \text{cis } \frac{2\pi}{3} \times \text{cis } \left(-\frac{\pi}{3}\right) \\ &= 2 \text{cis } \frac{\pi}{3} \quad \left[\text{as } \frac{2\pi}{3} + -\frac{\pi}{3} = \frac{\pi}{3}\right] \\ &= 2 \left[\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right] \\ &= 2 \left[\frac{1}{2} + i \frac{\sqrt{3}}{2} \right] \\ &= 1 + i\sqrt{3}\end{aligned}$$

e. i.



In $\triangle OXI$, by trig $\cos \frac{\theta}{2} = \frac{OI}{OR} = \frac{|OI|}{|OR|}$

$$\therefore z = |OR| \text{cis } \frac{\theta}{2}$$

$$\text{But } OR = 2 \times OX$$

$$\therefore OR = 2 \cos \frac{\theta}{2}$$

$$\therefore z = 2 \cos \frac{\theta}{2} (\cos \frac{\theta}{2} + i \sin \frac{\theta}{2})$$

$$\text{ii. } \frac{1}{z} = z^{-1} = [2 \cos \frac{\theta}{2} (\cos \frac{\theta}{2} + i \sin \frac{\theta}{2})]^{-1}$$

$$= \frac{1}{2 \cos \frac{\theta}{2}} [\cos -\frac{\theta}{2} + i \sin -\frac{\theta}{2}]$$

by de Moivre

$$= \frac{1}{2 \cos \frac{\theta}{2}} [\cos \frac{\theta}{2} - i \sin \frac{\theta}{2}]$$

$$= \frac{1}{2} \left[\frac{\cos \frac{\theta}{2} - i \sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} \right]$$

$$= \frac{1}{2} [1 - i \tan \frac{\theta}{2}]$$

$$= \frac{1}{2} - \frac{i}{2} \tan \frac{\theta}{2}$$

2

Question 3

a. i. Since w is a root of $z^3 = 1$
 α .

$$\therefore w^3 = 1$$

$$\beta. \therefore z^3 - 1 = 0$$

$$\therefore \text{sum of roots} = -\frac{b}{a} = 0$$

$$\therefore 1 + w + w^2 = 0$$

$$\text{ii. } x^3 + ax^2 + bx + c = 0$$

$$\therefore 1 + w + w^2 = -a$$

$$\therefore 0 = -a \quad (\text{from } \beta \text{ above})$$

$$\therefore a = 0$$

Sum of roots in pairs:

$$w + w^2 + w^3 = b$$

$$w(1 + w + w^2) = b$$

$$\therefore b = 0$$

Product of roots:

$$1(w)(w^2) = -c$$

$$w^3 = -c$$

$$\text{But } w^3 = 1 \therefore c = -1$$

$$\therefore a = 0, b = 0, c = -1$$

b. Diff wrt x :

$$3x^2 + 3y^2 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x^2}{y^2}$$

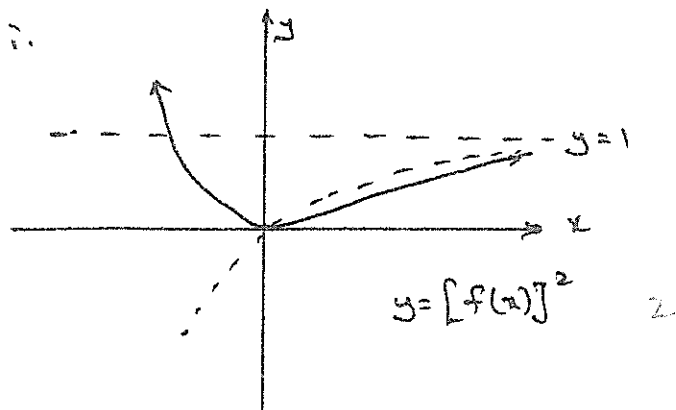
As $x=1, y=0$

$\therefore \frac{dy}{dx} = \text{undef.} \therefore \text{tangent vertical}$

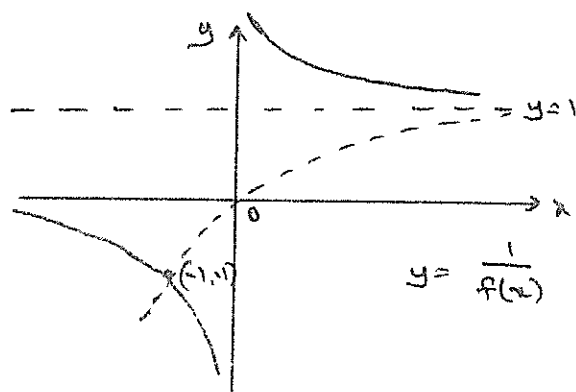
and for $y=x$, gradient $=1 \therefore \theta = 45^\circ$

$\therefore \angle \text{ between is } 45^\circ$

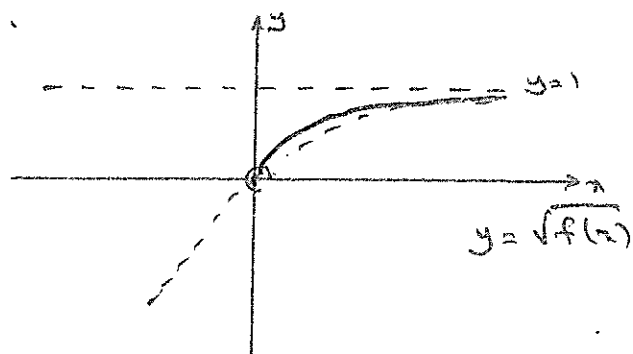
c. i.



ii.



iii.



Question 4

a. $z^{10} = 1 \therefore z=1$ is one root

Now, roots are equally spaced around unit circle

$\therefore \text{ spaced } \frac{2\pi}{10} \text{ apart ie } \frac{\pi}{5}$

$\therefore \text{ roots are } z = \cos \frac{r\pi}{5} + i \sin \frac{r\pi}{5}$

where $r=0, 1, 2, \dots, 9$ [$r=0 \rightarrow z=1$]

b. Roots of form $x = \frac{1}{\alpha}$ ie $\alpha = \frac{1}{x}$

$\therefore \text{ Subs } \frac{1}{x}$

$$\therefore \left(\frac{1}{x}\right)^3 + 2\left(\frac{1}{x}\right) + 1 = 0$$

$$\frac{1}{x^3} + \frac{2}{x} + 1 = 0$$

mult thru by x^3 :

$$1 + 2x^2 + x^3 = 0$$

$$\text{ie } x^3 + 2x^2 + 1 = 0$$

ii. As $x^3 + 2x + 1 = 0$ has roots

$$\alpha, \beta, \gamma \therefore \alpha + \beta + \gamma = 0$$

$$\therefore \beta + \gamma = -\alpha$$

$$\text{But if root is } \frac{\beta + \gamma}{\alpha^2} = \frac{-\alpha}{\alpha^2} = -\frac{1}{\alpha}$$

$\therefore \text{ question is roots are } \frac{1}{\alpha}, -\frac{1}{\beta}, -\frac{1}{\gamma}$

$\therefore \text{ equation is } \left(-\frac{1}{x}\right)^3 + \left(-\frac{2}{x}\right) + 1 = 0$

$$-1 - 2x^2 + x^3 = 0$$

$$\therefore x^3 - 2x^2 - 1 = 0$$

c. i. $\frac{dy}{d\theta} = 1 + \cos 2\theta$

$$\frac{dy}{d\theta} = 1 - \cos 2\theta$$

Now, $\frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx}$

$$= 1 - \cos 2\theta \times \frac{1}{1 + \cos 2\theta}$$

$$= \frac{1 - \cos 2\theta}{1 + \cos 2\theta}$$

$$= \frac{1 - (1 - 2\sin^2 \theta)}{1 + (2\cos^2 \theta - 1)}$$

$$= \frac{2\sin^2 \theta}{2\cos^2 \theta}$$

$$= \tan^2 \theta$$

ii. $\frac{d^2y}{dx^2} = \frac{d}{dx} [\tan^2 \theta]$

$$= \frac{d}{dx} [(\tan \theta)^2]$$

$$= 2 \tan \theta \sec^2 \theta \cdot \frac{d\theta}{dx}$$

$$= 2 \tan \theta \sec^2 \theta \cdot \frac{1}{1 + \cos 2\theta}$$

$$= 2 \tan \theta \sec^2 \theta \cdot \frac{1}{2\cos^2 \theta}$$

$$= \tan \theta \sec^4 \theta$$

d i. $x = 3 \cos \theta$

$$\frac{x}{3} = \cos \theta$$

$$\frac{x^2}{9} = \cos^2 \theta \quad \text{--- (1)}$$

Similarly, $\frac{y^2}{16} = \sin^2 \theta \quad \text{--- (2)}$

$$(2) + (1) \quad \frac{x^2}{9} + \frac{y^2}{16} = 1$$

ii. As $a > b$ always $\therefore a = 4, b = 3$

$\therefore$ for ellipse: $b^2 = a^2(1 - e^2)$

$$9 = 16(1 - e^2)$$

$$1 - e^2 = \frac{9}{16}$$

$$e^2 = \frac{7}{16}$$

$$e = \frac{\sqrt{7}}{4}$$

$\therefore$ foci $(0, \pm \sqrt{7})$

$\therefore S_1(0, \sqrt{7}), S_2(0, -\sqrt{7})$

iii. Directrices $y = \pm \frac{a}{e}$

$$\therefore y = \pm 4 \div \frac{\sqrt{7}}{4}$$

$$= \pm \frac{16}{\sqrt{7}}$$

$\therefore$ directrices $y = \frac{16}{\sqrt{7}}, y = -\frac{16}{\sqrt{7}}$

Question 5

a i. Let $x = a$ be the root of $P(x)$

$$\therefore P(x) = (x - a)^3 Q(x)$$

$$\therefore P'(x) = (x - a)^3 Q'(x) + Q(x) \cdot 3(x - a)^2$$

$$= (x - a)^2 [(x - a)Q'(x) + 3Q(x)]$$

which has a root of $x = a$ with multiplicity 2

ii. $P(x) = 8x^4 - 25x^3 + 27x^2 - 11x + 1$

$$P'(x) = 32x^3 - 75x^2 + 54x - 11$$

$$P''(x) = 96x^2 - 150x + 54 = 0$$

$$6(16x^2 - 25x + 9) = 0$$

$$6(16x - 9)(x - 1) = 0$$

$\therefore$ zeros of $P''(x)$ are $\frac{9}{16}, 1$

Now test in $P(x)$

$$P(1) = 8(1)^4 - 25(1)^3 + 27(1)^2 - 11(1) + 1 = 0$$

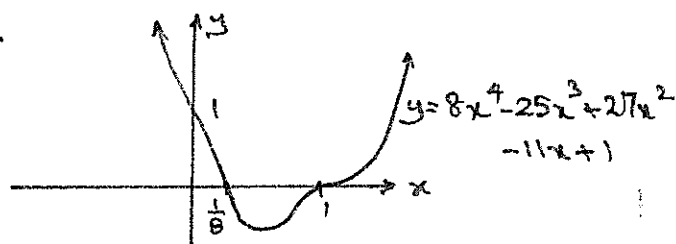
$\therefore x = 1$ is the triple root

$\therefore$ Sum of roots are $1 + 1 + 1 + \beta = \frac{25}{8}$

$$\beta = \frac{25}{8} - 3 = \frac{1}{8}$$

$\therefore$ zeros are $1, 1, 1, \frac{1}{8}$

iii.



b. $a \tan^2 d + b \tan d + c = 0$

Roots are $\tan \alpha_1, \tan \alpha_2$

$$\therefore \tan \alpha_1 + \tan \alpha_2 = -\frac{b}{a}$$

$$\tan \alpha_1 \tan \alpha_2 = \frac{c}{a}$$

$$\therefore \tan(\alpha_1 + \alpha_2) = \frac{\tan \alpha_1 + \tan \alpha_2}{1 - \tan \alpha_1 \tan \alpha_2}$$

$$= \frac{-\frac{b}{a}}{1 - \frac{c}{a}}$$

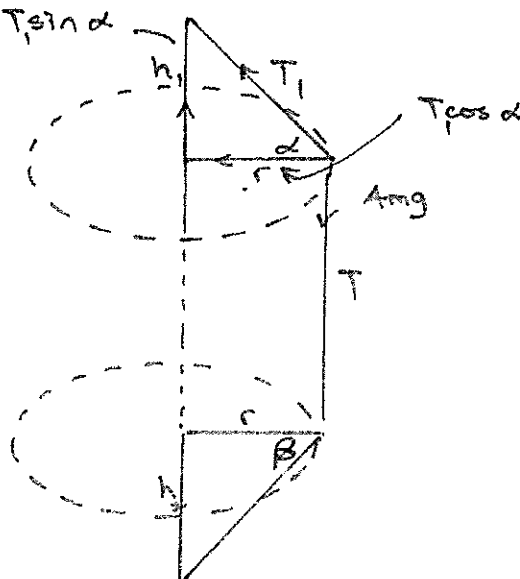
$$= \frac{-\frac{b}{a}}{\frac{a-c}{a}}$$

$$= \frac{-\frac{b}{a} \times a}{a-c}$$

$$= \frac{b}{c-a}$$

c. acceleration is rw^2 and directed towards centre of the circle

d. $T \sin \alpha$



Resolving vertically: $T \sin \alpha - 4mg - T = 0$

$$\therefore T \sin \alpha = 4mg + T \quad \text{--- (1)}$$

Resolving horizontally:

$$T_1 \cos \alpha = (4m) r \omega^2$$

$$\therefore T_1 \cos \alpha = 4mr\omega^2 \quad \text{--- ②}$$

ii. ① $\div$ ②

$$\frac{T \sin \alpha}{T \cos \alpha} = \frac{4mg + T}{4mr\omega^2}$$

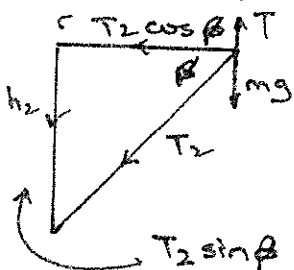
$$\tan \alpha = \frac{4mg + T}{4mr\omega^2}$$

Now, by trig: $\tan \alpha = \frac{h_1}{r}$

$$\therefore \frac{h_1}{r} = \frac{4mg + T}{4mr\omega^2}$$

$$\therefore h_1 = \frac{4mg + T}{4m\omega^2}$$

iii. Consider forces at Q



Let tension be T_2

Vertically: $T - T_2 \sin \beta - mg = 0$

$$\therefore T_2 \sin \beta = T - mg \quad \text{--- ①}$$

Horizontally:

$$T_2 \cos \beta = mr\omega^2 \quad \text{--- ②}$$

$$\text{①} \div \text{②} \quad \tan \beta = \frac{T - mg}{mr\omega^2}$$

From trig: $\tan \beta = \frac{h_2}{r}$

$$\therefore \frac{h_2}{r} = \frac{T - mg}{mr\omega^2}$$

$$\therefore h_2 = \frac{T - mg}{m\omega^2}$$

Now $(4h_1 - h_2)\omega^2$

$$= \left(\frac{4mg + T}{m\omega^2} - \frac{T - mg}{m\omega^2} \right) \omega^2$$

$$= \frac{4mg + T - T + mg}{m}$$

$$= 5g$$

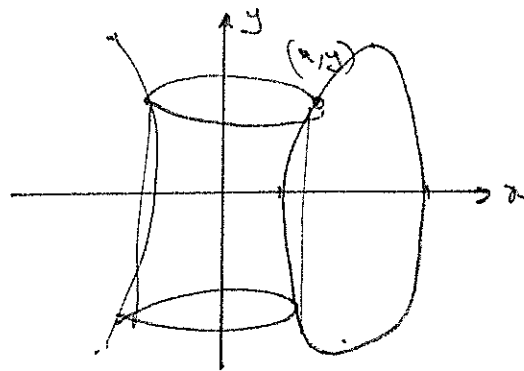
Question 6

a. $P(x) = Q(x)(x+2)(x-3) + (4x+1)$

$$\therefore P(-2) = 0 + (4(-2) + 1) = -7$$

$\therefore$ remainder is -7

b.



i. $V = 2\pi \int r h \, dx$

$$= 2\pi \int_1^3 x \cdot 2y \, dx$$

$$= 4\pi \int_1^3 xy \, dx$$

Now, $(x-2)^2 + \frac{y^2}{4} = 1$

$$\frac{y^2}{4} = 1 - (x-2)^2$$

$$y^2 = 4[1 - (x-2)^2]$$

$$y = 2\sqrt{1 - (x-2)^2}$$

$$\therefore V = 8\pi \int_1^3 x \sqrt{1 - (x-2)^2} \, dx$$

ii. Let $u = x - 2$

$$\therefore \frac{du}{dx} = 1$$

$$dx = du$$

$$\therefore V = 8\pi \int_{-1}^1 (u+2) \sqrt{1-u^2} \, du$$

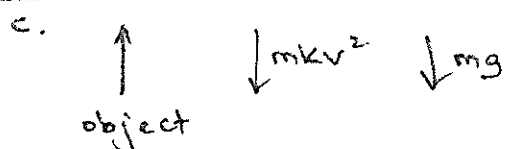
$$= 8\pi \left[\int_{-1}^1 u \sqrt{1-u^2} \, du + \int_{-1}^1 2\sqrt{1-u^2} \, du \right]$$

Now, $u\sqrt{1-u^2}$ is an odd function

$$\therefore \int u \sqrt{1-u^2} \, dx = 0$$

and $\int_{-1}^1 \sqrt{1-u^2} \, du$ is area of semi-circle, radius 1

$$\therefore 8\pi \left[0 + 2 \left(\frac{1}{2} \pi \times 1^2 \right) \right] = 8\pi$$



i. $m\ddot{x} = -mkv^2 - mg$

$\therefore \ddot{x} = -kv^2 - g$

$\therefore v \frac{dv}{dx} = -(g + kv^2)$

$\frac{dv}{dx} = \frac{-(g + kv^2)}{v}$

$\frac{dv}{v} = \frac{-v}{g + kv^2}$

$x = -\frac{1}{2k} \ln(g + kv^2) + c$

$x=0, v=U \therefore 0 = -\frac{1}{2k} \ln(g + kU^2) + c$

$c = \frac{1}{2k} \ln(g + kU^2)$

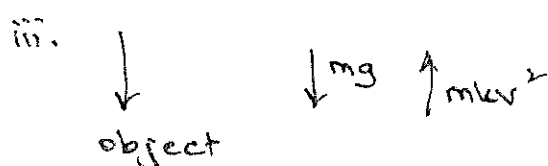
$\therefore x = \frac{1}{2k} \ln\left(\frac{g + kU^2}{g + kv^2}\right) \quad \text{--- ①}$

Now, $x=H, v=0$ $H = \frac{1}{2k} \ln\left(\frac{g + kU^2}{g}\right)$

ii. At P, $x=h, v=V$

Subs in ①

$h = \frac{1}{2k} \ln\left(\frac{g + kU^2}{g + kV^2}\right)$



$m\ddot{x} = mg - mkv^2$

$\ddot{x} = g - kv^2$

$v \frac{dv}{dx} = g - kv^2$

$\frac{dv}{dx} = \frac{g - kv^2}{v}$

$\frac{dv}{v} = \frac{v}{g - kv^2}$

$x = -\frac{1}{2k} \ln(g - kv^2) + c$

Let starting point by $x=0$.

$\therefore x=0, v=0 \quad 0 = -\frac{1}{2k} \ln(g - 0) + c$

$c = \frac{1}{2k} \ln g$

$\therefore x = \frac{1}{2k} \ln\left(\frac{g}{g - kv^2}\right)$

Let h be distance from ground

$\therefore$ fallen $H-h$

$\therefore x = H-h, v = \frac{1}{2}V$

$\therefore H-h = \frac{1}{2k} \ln\left(\frac{g}{g - k\left(\frac{V}{2}\right)^2}\right)$
 $= \frac{1}{2k} \ln\left(\frac{4g}{4g - kV^2}\right)$

Now, using H from i and h from ii

$\frac{1}{2k} \ln\left(\frac{g + kU^2}{g}\right) - \frac{1}{2k} \ln\left(\frac{g + kU^2}{g + kV^2}\right)$
 $= \frac{1}{2k} \ln\left(\frac{4g}{4g - kV^2}\right)$

mult thru by $2k$:

$\ln\left(\frac{g + kU^2}{g} \times \frac{g + kV^2}{g + kU^2}\right)$
 $= \ln \frac{4g}{4g - kV^2}$

$\therefore \frac{g + kV^2}{g} = \frac{4g}{4g - kV^2}$

$4g - kV^2 + 4gV^2 - k^2V^4 = 4g^2$
 $k^2V^4 = 3kV^2$

$V^2 = \frac{3g}{k}$

$\therefore V = \sqrt{\frac{3g}{k}}$

Question 7

a. $z = a + ib$

$\therefore a^2 + b^2 + 5(a - ib) + 10i = 0 + 0i$

$\therefore a^2 + b^2 + 5a = 0 \quad \text{--- ①}$

$-5b + 10 = 0 \quad \text{--- ②}$

$$\therefore b=2$$

Subs in ① $a^2 + 4 + 5a = 0$

$$a^2 + 5a + 4 = 0$$

$$(a+4)(a+1) = 0$$

$$a = -4, -1$$

$$\therefore -4+2i, -1+2i$$

b. i. $xy = 4$

$\therefore$ Diff wrt x :

$$y + x \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

At P: grad = $-\frac{y}{x} \times \frac{1}{x}$

$$= -\frac{1}{p^2}$$

ii. grad of normal = p^2

$$\therefore y - \frac{y}{p} = p^2(x - 2p)$$

$$py - 2 = p^3(x - 2p)$$

$$py - 2 = p^3x - 2p^4$$

$$\therefore p^3x - py = 2(p^4 - 1) \quad \text{--- ①}$$

iii. As $xy = 4 \therefore y = \frac{4}{x}$

Subs in ①

$$p^3x - \frac{4p}{x} = 2(p^4 - 1)$$

$$p^3x^2 - 4p = 2x(p^4 - 1)$$

$$p^3x^2 - 4p - 2p^4x + 2x = 0$$

$$\therefore p^3x^2 - 2(p^4 - 1)x - 4p = 0$$

Now, product of roots = $-\frac{4p}{p^3} = -\frac{4}{p^2}$

But roots are $2p$ and $2q$

$$\therefore 4pq = -\frac{4}{p^2}$$

$$\therefore p^3q = -1$$

c. RHS = $t^{n-2} - \frac{t^{n-2}}{1+t^2}$

$$= \frac{(1+t^2)t^{n-2} - t^{n-2}}{1+t^2}$$

$$= \frac{t^{n-2} + t^n - t^{n-2}}{1+t^2}$$

$$= \frac{t^n}{1+t^2}$$

= LHS

ii. $I_n = \int \frac{t^n}{1+t^2} dt$

$$= \int t^{n-2} - \frac{t^{n-2}}{1+t^2} dt$$

$$= \frac{t^{n-1}}{n-1} - \int \frac{t^{n-2}}{1+t^2} dt$$

$$= \frac{t^{n-1}}{n-1} - I_{n-2}$$

iii. let $J_n = \int_0^1 \frac{t^n}{1+t^2} dt$

$$= \frac{t^{n-1}}{n-1} \Big|_0^1 - J_{n-2}$$

$$= \frac{1}{n-1} - J_{n-2}$$

$$\therefore J_6 = \frac{1}{5} - J_4$$

$$= \frac{1}{5} - \left[\frac{1}{3} - J_2 \right]$$

$$= \frac{1}{5} - \frac{1}{3} + 1 - J_0$$

But $J_0 = \int_0^1 \frac{1}{1+t^2} dt$

$$= \tan^{-1} \Big|_0^1$$

$$= \pi/4 - 0$$

$$= \pi/4$$

$$\therefore J_6 = \frac{1}{5} - \frac{1}{3} + 1 - \pi/4$$

$$= \frac{13}{15} - \frac{\pi}{4}$$

Question 8

a. $y = e^{xy}$

$$\frac{dy}{dx} = \left(y + x \frac{dy}{dx} \right) e^{xy}$$

$$\therefore \frac{dy}{dx} = ye^{xy} + x \frac{dy}{dx} e^{xy}$$

$$\frac{dy}{dx} (1 - xe^{xy}) = ye^{xy}$$

$$\therefore \frac{dy}{dx} = \frac{ye^{xy}}{1 - xe^{xy}}$$

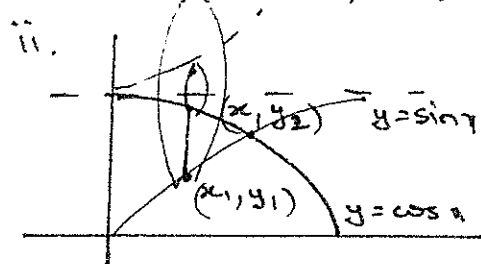
b. i. $\sin x = \cos x$

$$\therefore \frac{\sin x}{\cos x} = \frac{\cos x}{\cos x}$$

$$\tan x = 1$$

$$x = \pi/4$$

$$\therefore (\pi/4, \frac{1}{\sqrt{2}})$$



$$\begin{aligned} \delta V &= \pi (\text{radii of annulus squared}) \delta x \\ &= \pi ([1 - \sin x]^2 - [1 - \cos x]^2) \delta x \\ &= \pi (1 - 2\sin x + \sin^2 x - 1 + 2\cos x - \cos^2 x) \delta x \\ &= \pi (2\cos x - 2\sin x + \sin^2 x - \cos^2 x) \delta x \end{aligned}$$

$$\begin{aligned} \text{iii. } V &= \pi \int_0^{\pi/4} (2\cos x - 2\sin x - \cos 2x) dx \\ &= \pi [2\sin x + 2\cos x - \frac{1}{2}\sin 2x]_0^{\pi/4} \\ &= \pi [2 \cdot \frac{1}{\sqrt{2}} + 2 \cdot \frac{1}{\sqrt{2}} - \frac{1}{2} - (0 + 2 - 0)] \\ &= \pi [2\sqrt{2} - \frac{5}{2}] \\ &= \frac{\pi}{2} [4\sqrt{2} - 5] \text{ units}^3 \end{aligned}$$

c. i. Now tangent AC:

$$y = e^x \quad \therefore y' = e^x$$

$$y'(0) = e^0 = 1$$

$$\therefore \text{eqn: } y - 1 = 1(x - 0)$$

$$y - 1 = x$$

$$\therefore y = x + 1 \quad \text{--- (1)}$$

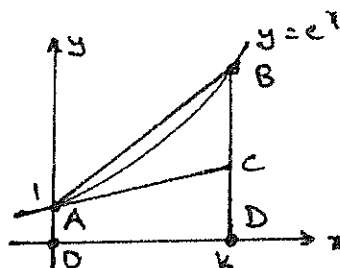
Now subs $x = k$ in (1) $\therefore y = k + 1$

$$\therefore C(k, k+1)$$

Also, $B(k, e^k)$

Now, from diagram:

$$\text{Area AODC} < \int_0^k e^x dx < \text{Area AODB}$$



$$\therefore \frac{1}{2}k(1+k+1) < e^x \Big|_0^k < \frac{1}{2}k(1+e^k)$$

$$\frac{1}{2}k(k+2) < e^k - 1 < \frac{1}{2}k(1+e^k)$$

Now, let $k=1$

$$\therefore 1.5 < e - 1 < \frac{1+e}{2}$$

$$\begin{aligned} \therefore e - 1 &> 1.5 & 2e - 2 &< 1 + e \\ e &> 2.5 & e &< 3 \end{aligned}$$

$$\therefore 2.5 < e < 3$$

ii. If area $\triangle ABC$ is bisected

$$\therefore e^k - 1 - \frac{1}{2}k(k+2) = \frac{1}{2}k(1+e^k) - (e^k - 1)$$

$$2e^k - 2 - k^2 - 2k = k + ke^k - 2e^k + 2$$

$$4e^k - ke^k - k^2 - 3k - 4 = 0$$

$$(4-k)e^k - k^2 - 3k - 4 = 0$$

$$\text{Let } f(k) = (4-k)e^k - k^2 - 3k - 4$$

$$\text{Now } f(2) = 0.78 > 0$$

$$f(3) = -1.9 < 0$$

$\therefore$ as $f(k)$ is continuous,

$\therefore k$ lies between 2 & 3 if

$$f(k) = 0 \quad \therefore 2 < k < 3$$

$$\begin{aligned} \text{Now } f'(k) &= -e^k + (4-k)e^k - 2k - 3 \\ &= (3-k)e^k - 2k - 3 \end{aligned}$$

$$f'(2.7) = 3.9361 \quad f(2.7) = -0.4635$$

$$\therefore k_1 = 2.7 - \frac{-0.4635}{3.9361}$$

$$= 2.688$$

$\therefore$ 2nd approx is 2.7 (1 dp)